

The question arises again, how did they manage to do it?

The solution is quite innovative, actually. The system consists of three main units:

The Reservation Station: This is where the graphics instructions are saved. Now the reservation system can store either pixel or vertex shader commands.

The Arbiter: The Arbiter sits in Olympian judgment over the Reservation Station and takes and 'intelligently' transfers them to the Command Processing Engine.

The Command Processing Engine: It receives commands from the Arbiter, and after execution, sends them back to the Reservation Station—and includes some form of a flag system that marks the command as 'completed' so as to avoid multiple execution.

Now, to complicate things, instead of having a generic Reservation Station, let's put two Reservation Stations, where one stores vertex commands and the other stores pixel commands. These are directly connected to the Arbiter, which in turn is connected to the ALU and a Texture Processing Unit.

The Arbiter can now choose whether a command from the Reservation Stations goes to the ALU or the Texture Processing Unit, and the processed results are then sent back to the appropriate Reservation Station.

All this processing power and the spanking new graphics core implantation from ATI will ensure that the Xbox 360 stays competitive to the PCs and other consoles for the regular life of a console—around five years.

Sony and nVidia have been mum on the Graphics engine used in the PS3 as it has not yet been launched—it's expected in 2006—so we can't talk about it here.

The Consoles

The consoles of today are morphing from pure gaming devices into complete entertainment hubs with the ability to play DVDs, music and pictures. Let's take them apart feature by feature and see what's on offer.

The Xbox 360

The 360 is called so for the single reason that Microsoft thought calling it 'Xbox 2' would make it seem outdated in front of a name like 'PS3' (2 comes *before* 3, right?)

But the name is apt, as it does provide for a 360-degree entertainment experience. If you buy an Xbox, it will work as your music centre, DVD Player (12x dual-layer), as well as a photo sharer.

It comes with an inbuilt network adapter that lets you chat (text and voice) with other 360 users, whether they're watching a movie or smoking fellow racers in NFS Most Wanted, apart from connecting to your home network and surfing the Internet wirelessly!

The device also has support for up to four wireless controllers, and has three USB2.0 ports. An added attraction is the changeable face plate (à la Nokia) which will let you customise your Xbox 360 to match your décor (it is definitely sleek enough to be put there!) and taste.

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The Controllers

Things haven't changed too much (not counting the pretty designs). The face is devoid of the ugly black and white buttons, which have been relocated onto the shoulder. Both wired and wire-free controls will be available. The wirefree versions will be powered by 2 AA batteries, which you can recharge via a cable. The central Xbox Guide button gives you instant access to your favourite functions such as movies, music or contacting friends over Xbox Live.

Xbox Live!

Xbox Live, the online service from Microsoft, is now available in two avatars—Gold and Silver. Each Xbox will come with a free subscription to the Silver version, which allows players to share their gamer's profiles (consisting of stuff like races won) and a customisable gamerstyle



The new Xbox controller is a lot sleeker than its predecessor

(marketing speak for display image!). It will also allow you to chat (text/voice) and log on to Live Marketplace, where you can download demos, trailers, new levels/maps etc. and store them on the detachable 20 GB hard disk.

Gold subscribers get the added benefit of online multiplayer gameplay, as well as video messaging. All Xbox 360 owners who sign on to Xbox Live for the first time will get a free month of the full subscription service, which will let them experience the full power of Live. Those who currently use Live on Xbox will also be able to transfer their subscriptions straight over to Gold.



The Xbox 360 Gamer's Profile window

In terms of availability, the Xbox 360 has been showcased, but Microsoft has not yet started shipping. The expected price would be somewhere around \$300, or roughly Rs 13,500.

The previous version is, of course, available in India, and you can pick up an Xbox (not an